EXP 2: Simulation and Performance Evaluation of People–Process–Data–Things IoE Framework using Edge/Fog-to-Cloud Pipeline

Objective:

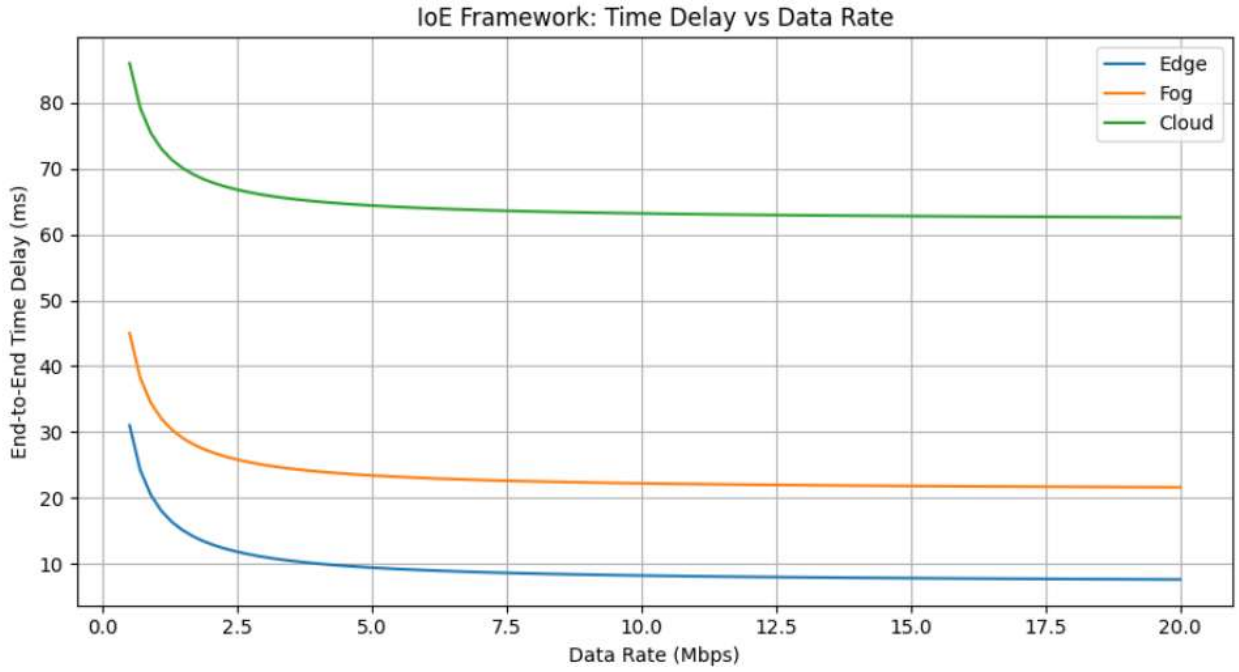
To simulate and evaluate a People–Process–Data–Things (PPDT) Internet of Everything (IoE) framework using an Edge/Fog-to-Cloud pipeline to improve data processing efficiency, reduce latency, optimize bandwidth usage, and enhance system performance, reliability, and scalability.

Procedure:

1. Design the PPDT-based IoE framework.
2. Simulate IoE devices generating real-time data.
3. Process data at the Edge and Fog layers.
4. Transfer processed data to the Cloud for storage and analysis.
5. Evaluate latency, throughput, bandwidth, and energy efficiency.
6. Compare the results with a cloud-only approach.
7. Analyze the results to assess overall system performance.

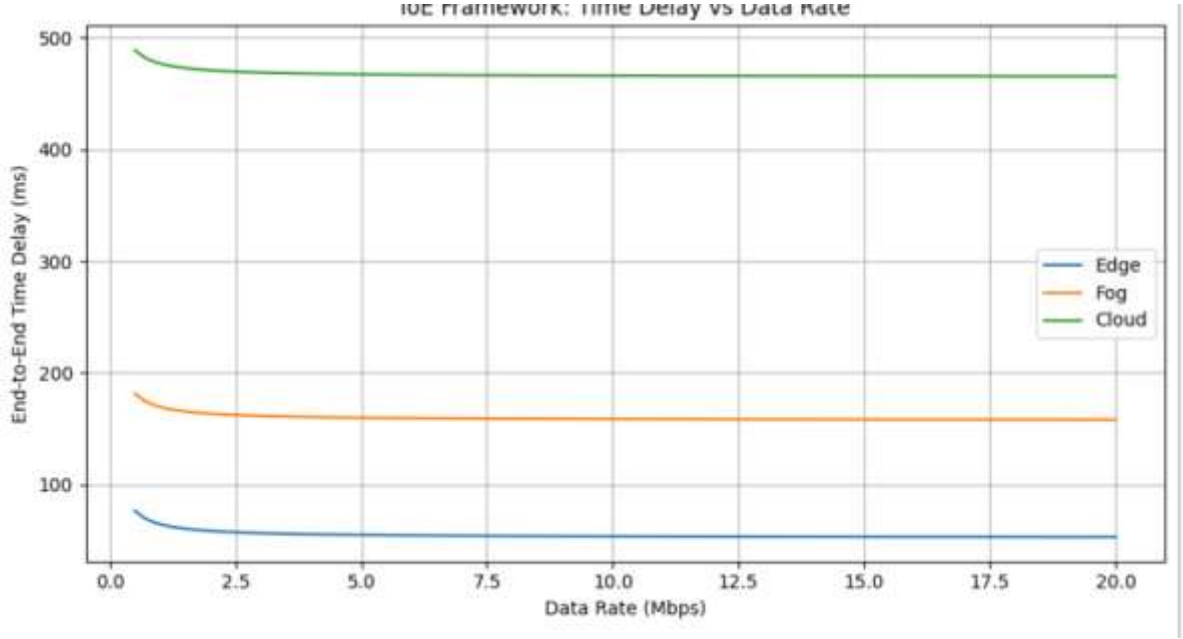
Number of sources: 2

	Edge	Fog	Cloud
Propagation delay	2	10	40
Processing delay	4	8	16
Queueing delay	1	3	6
End to end	8.159ms	22.15ms	63.15ms



Number of source:15

	Edge	Fog	Cloud
Propagation delay	15	75	300
Processing delay	30	60	120
Queueing delay	7.5	22.5	45
End to end	53.659ms	158.659 ms	466.159



Result:

Edge computing provides the lowest end-to-end delay compared to Fog and Cloud for both 2 and 15 sources. As the number of sources increases, delay also increases. Therefore, Edge is best for real-time IoE applications, Fog provides moderate performance, and Cloud is suitable for large-scale data processing.